FLAIR Task Schematic

Sepsis Prediction with Antibiotic-Anchored Time-Zero

FLAIR Benchmark

June 4, 2026

Abstract

We define a new FLAIR benchmark task that predicts *imminent first qualifying intravenous antibiotic administration* as an operational proxy for clinician recognition of sepsis. The task fixes time-zero (t_{zero}) at the first qualifying antibiotic event, starts predictions at the first recorded vital sign, and stops predictions strictly before t_{zero} to avoid the label-leakage failure mode documented by Kamran et al. (2024). This document surveys ten prior operationalizations of sepsis onset, motivates the FLAIR choice, and specifies the labeling, cohort, and censoring rules with TikZ schematics. Input feature design is out of scope.

1 Introduction

Sepsis prediction is the most-studied AI use case in inpatient EHR research, yet there is no consensus on *when* a model's prediction should be considered correct (Singer et al., 2016; Rhee et al., 2019). The choice of time-zero changes both the label and the cohort, and Kamran et al. (2024) recently showed that the Epic Sepsis Model's AUROC drops from 0.62 to 0.47 when predictions made *after* a clinician's first sepsis-treatment order are excluded. This is the *label-leakage* problem: when the gold label is itself an administrative event (the antibiotic order), any prediction made after that order trivially correlates with the outcome it is supposed to forecast.

FLAIR's existing tasks (intubation, AKI, mortality, readmission) use clinical events that are either clearly involuntary (deterioration, death) or that occur at a clean state transition (ICU discharge). Sepsis is fundamentally different: the label is a clinician action. The task therefore must (i) pick an unambiguous anchor and (ii) censor predictions made at or after that anchor.

We adopt the antibiotic-anchored framework: t_{zero} is the first qualifying systemic IV antibiotic, and the prediction window terminates strictly before t_{zero}. This mirrors the evaluation protocol of Kamran et al. (2024) translated into a *training label* rather than a post-hoc evaluation gate.

2 Survey of prior time-zero definitions

Table 1 summarizes ten prior definitions and what each implies for prediction-window start, stop, and label leakage.

Table 1: Operational time-zero definitions across major sepsis-prediction studies.

Paper / model	Time-zero rule	Prediction horizon	Stop rule
Seymour et al. (2016) Sepsis-3 (JAMA)	t_{susp} = earlier of {IV abx, blood culture} provided abx→cult $\leq$ 24h or cult→abx $\leq$ 72h; $t_{\text{sepsis}} = t_{\text{susp}} + (\Delta\text{SOFA} \geq 2$ in $(-48, +24)$h)	— (definition paper)	—
Reyna et al. (2020) PhysioNet/CinC 2019	Seymour rule + IV abx sustained $\geq$72h; t_{sepsis} = earlier of $t_{\text{susp}}, t_{\text{SOFA}}$ if $t_{\text{SOFA}} \in (-24, +12)$h of t_{susp}	6h ahead, hourly	onset or discharge
Desautels et al. (2016) InSight	t_{susp} = earliest culture or abx; SOFA window $(-48, +24)$h	hourly, multi-horizon	onset
Shashikumar et al. (2021a) DeepAISE	Seymour rule; exclude sepsis in first 4h of ICU; ICU LOS 8h–20d	$T \in \{4, 6, 8, 12\}$h ahead	onset / discharge
Shashikumar et al. (2021b) COMPOSER	blood culture + $\geq$ 4d non-prophylactic IV abx, Seymour pairing	continuous	onset
Epic ESM(Wong et al., 2021)	proprietary; evaluated against Sepsis-3	every 20 min throughout stay	discharge
Adams et al. (2022) TREWS	Sepsis-3 with provider confirmation	hourly until first abx or provider override	provider confirms / abx
Kamran et al. (2024) (user-cited)	**Label** = composite of CDC ASE (Rhee et al., 2019) and CMS SEP-1 (Centers for Medicare and Medicaid Services, 2015). **Evaluation gate**: predictions are restricted to time-points strictly *before* the first "treatment indicator" timestamp $\in$ {antibiotics, blood culture, lactate, IV fluids}. *No fixed Δ-gate* — the indicator timestamp itself is the cutoff.	ESM scores every 20 min throughout admission	min(first treatment indicator, sepsis criteria met, death, discharge)
Rhee et al. (2019) CDC ASE	$\geq$ 4d qualifying abx within $\pm$2d of blood culture + eSOFA organ dysfunction	calendar-day	—
Centers for Medicare and Medicaid Services (2015) CMS SEP-1	2 SIRS criteria + 1 organ dysfunction within 6h of one another	—	—

Implications for FLAIR. Three patterns emerge from Table 1. First, the Seymour-paired definition dominates the AI-sepsis literature but is sensitive to culture order timing, which itself reflects clinician suspicion. Second, no two studies use the same cohort entry rule: ICU-only (Desautels et al., 2016; Shashikumar et al., 2021a) vs. hospital-wide (Kamran et al., 2024; Wong et al., 2021) vs. ED-onset (Henry et al., 2022) produce materially different prevalence estimates. Third, Kamran et al. (2024) is the first study to *quantify* the cost of label leakage by re-evaluating an existing deployed model under a stricter "before clinician action" rule. We adopt their pre-treatment-indicator gate as a structural property of the training label itself.

3 FLAIR operational definition

3.1 Anchors and windows

Let H_i denote a hospitalization with admission time t_i^{adm} and discharge or death time t_i^{end}.

Time-zero (t_{zero}).
$t^i_{\text{zero}} = \min\{\, t \mid \text{med_admin}(i,t) \in \mathcal{A}_{\text{IV-systemic}} \,\}$ where $\mathcal{A}_{\text{IV-systemic}}$ is the curated systemic-IV-antibiotic drug list (excludes topicals, prophylaxis single-doses, oral abx).

Prediction-window start (pred_start).
$\text{pred_start}_i = \min\{\, t \mid \text{vitals}(i,t) \neq \emptyset \,\}$ — the first recorded vital sign per hospitalization. Floor at $t^{\text{adm}}_i + 1\text{h}$ to guarantee a non-degenerate window when vitals are taken immediately at admission.

Prediction-window stop (pred_stop).
$\text{pred_stop}_i = \min(t^i_{\text{zero}},\ t^{\text{end}}_i)$. The interval $[\text{pred_start}_i, \text{pred_stop}_i)$ is closed-open; t^i_{zero} itself is *not* a decision point.

Decision-point grid.
Hourly grid: $D_i = \{\text{pred_start}_i + k\cdot 1\text{h} \ :\ k \in \mathbb{Z}_{\geq 0},\ \text{pred_start}_i + k\cdot 1\text{h} < \text{pred_stop}_i\}$.

Label (`label_sepsis_abx_6h`).
For $d \in D_i$:

$$y_i(d) = \mathbb{1}\Big[t^i_{\text{zero}} \in (d,\ d + 6\text{h}\,]\Big].$$

Patients with no qualifying antibiotic ever ($t^i_{\text{zero}} = +\infty$) contribute all-negative rows over the entire $[\text{pred_start}_i, t^{\text{end}}_i)$.

3.2 Cohort

Hospital-wide, mirroring Kamran et al. (2024):

- Age at admission ≥ 18 with valid `admission_dttm` and `discharge_dttm`.
- Exclude hospitalizations on psychiatric or rehabilitation units (Kamran 2024, Methods).
- Exclude $t^i_{\text{zero}} \leq \text{pred_start}_i$ (admitted-with-treatment; Kamran 2024: "patients ... who received treatment before the first measurement of vital signs").
- Exclude hospitalizations with no vital-sign measurements (cannot anchor pred_start).

3.3 CLIF tables and categories used

Task T4 reads six CLIF v2.1 tables — the largest CLIF surface of any FLAIR task. Note the structural asymmetry with Tasks T1, T2, T3, and T5: T4 is the only task whose ADT filter *excludes by category* rather than restricting to `"icu"`. This reflects the hospital-wide cohort (Kamran et al., 2024), which captures ward-onset sepsis where prediction lead time matters most.

CLIF table	Category column	Literal values kept (case-
hospitalization	—	demographics join; no catego
patient	—	demographics join; no catego
adt	location_category	*exclude* {"psychiatric", "re all other locations kept
vitals	vital_category	none (vital_categories=No to derive pred_start from the tamp
medication_admin_intermittent	med_group/med_route_category/mar_action_group†	med_group "cms_sepsis_qualifying_a ∧ med_route_category mar_action_group = "admi
medication_admin_continuous	med_group/med_route_category	med_group "cms_sepsis_qualifying_a med_route_category = "iv

Table 2: CLIF surface for Task T4. The exclusion-style ADT filter is unique to T4; a site whose location_category vocabulary lacks the literal "psychiatric" or "rehabilitation" will *silently retain those stays*, unlike Tasks T1/T2/T3/T5 where a mapping gap would zero out the cohort loudly.

† The mar_action_group predicate is applied with a defensive conditional: if the column is absent from a site's medication_admin_intermittent export, the filter is skipped silently rather than dropping every row. The continuous-infusion table does not carry this filter at all because the relevant administrative state is encoded in the start/stop semantics of the continuous infusion itself.

Columns selected from each table:

- hospitalization: hospitalization_id, patient_id, admission_dttm, discharge_dttm, age_at_admission, discharge_category.
- patient: patient_id, sex_category, race_category, ethnicity_category, death_dttm.
- adt: hospitalization_id, location_category, in_dttm, out_dttm.
- vitals: hospitalization_id, recorded_dttm.
- medication_admin_intermittent: hospitalization_id, admin_dttm.
- medication_admin_continuous: hospitalization_id, admin_dttm.

The antibiotic drug list (med_group = "cms_sepsis_qualifying_antibiotics") is the largest open dependency for this task; see §6, *Drug list* for the curation rationale.

3.4 Sensitivity arms (documentation only, not primary output)

- **ICU-only arm.** Restrict to hospitalizations with ≥ 1 ICU ADT record; pred_start$_i$ = max(first_icu_start$_i$ + 1h, first vital).
- **Seymour-paired arm.** Require blood culture in (t_{zero} − 72h, t_{zero} + 24h) for the antibiotic to qualify.

- **Sepsis-3 label arm.** Replace abx-as-label with $\Delta\text{SOFA} \geq 2$ within ± 48h of t_{susp} (PhysioNet 2019 rule) for direct cross-comparison.

4 Algorithmic flow

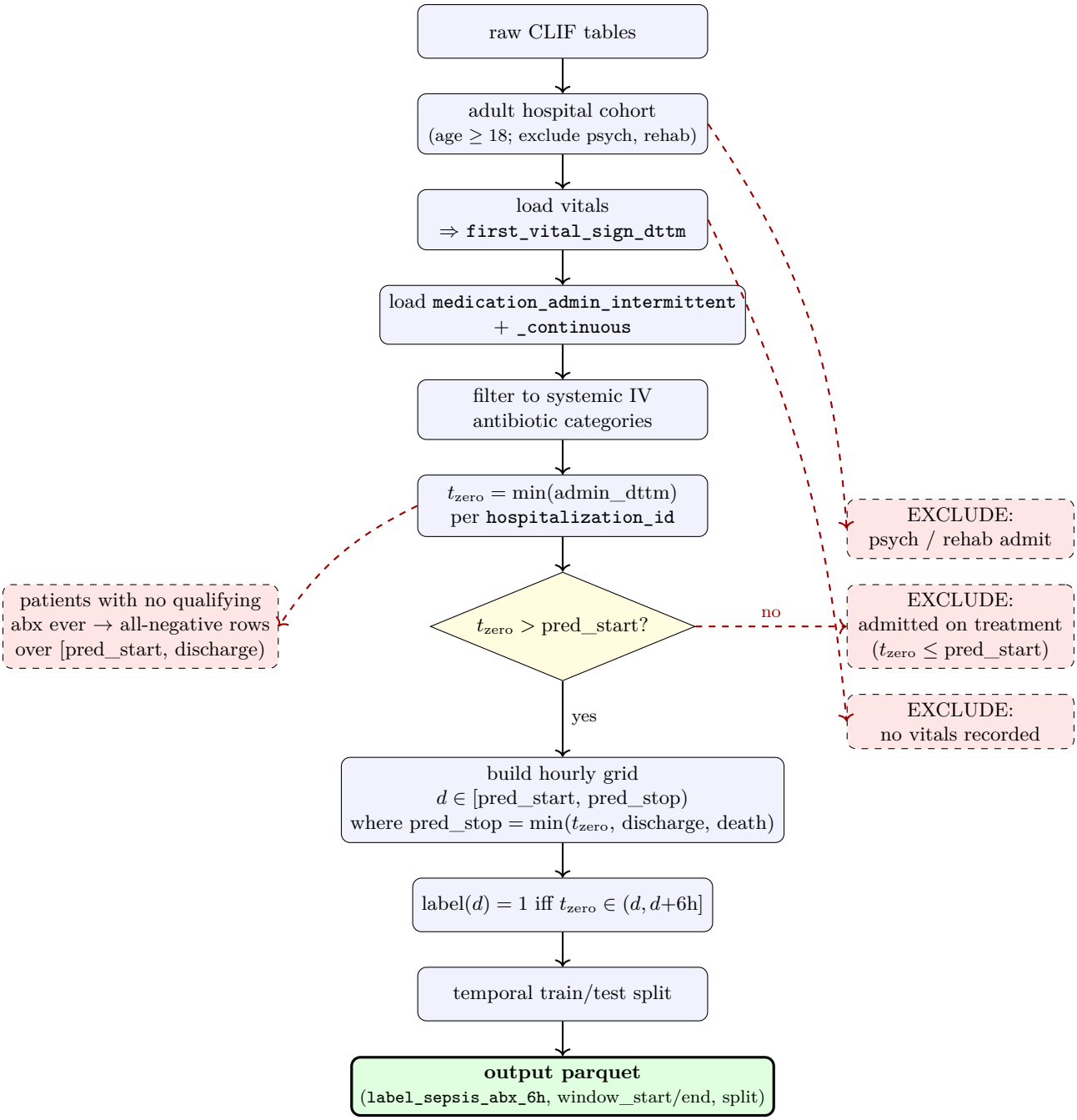


Figure 1: Build pipeline. Solid arrows: main flow. Dashed red arrows: exclusions. The diamond enforces the label-leakage gate: any hospitalization whose qualifying antibiotic precedes its first vital sign is dropped, since by Kamran et al.'s argument no decision point in that stay can produce a recognition-free prediction.

Step-by-step prose:

1. Load CLIF tables: `hospitalization`, `patient`, `adt`, `vitals`, and both `medication_admin_intermittent` / `medication_admin_continuous`.
2. Apply hospital-wide cohort filters: age, psych/rehab exclusion.
3. Compute pred_start$_i$ from `vitals.recorded_dttm`.
4. Filter `medication_admin_*` to $\mathcal{A}_{\text{IV-systemic}}$; compute $t^i_{\text{zero}} = \min(\text{admin_dttm})$ per `hospitalization_id`.
5. Apply the leakage gate: drop $t^i_{\text{zero}} \leq \text{pred_start}_i$.
6. Explode the hourly decision-point grid [pred_start$_i$, pred_stop$_i$) per hospitalization (`pl.datetime_ranges`, closed-open).
7. Assign $y_i(d)$ per the label rule.
8. Assign `window_start` $= d - 24$h, `window_end` $= d$ (lookback window, fixed 24h to match the FLAIR convention).
9. Apply temporal train/test split per `flair_benchmark._split.assign_split`.
10. Select `OUTPUT_COLS` and persist as parquet.

5 Timeline diagram

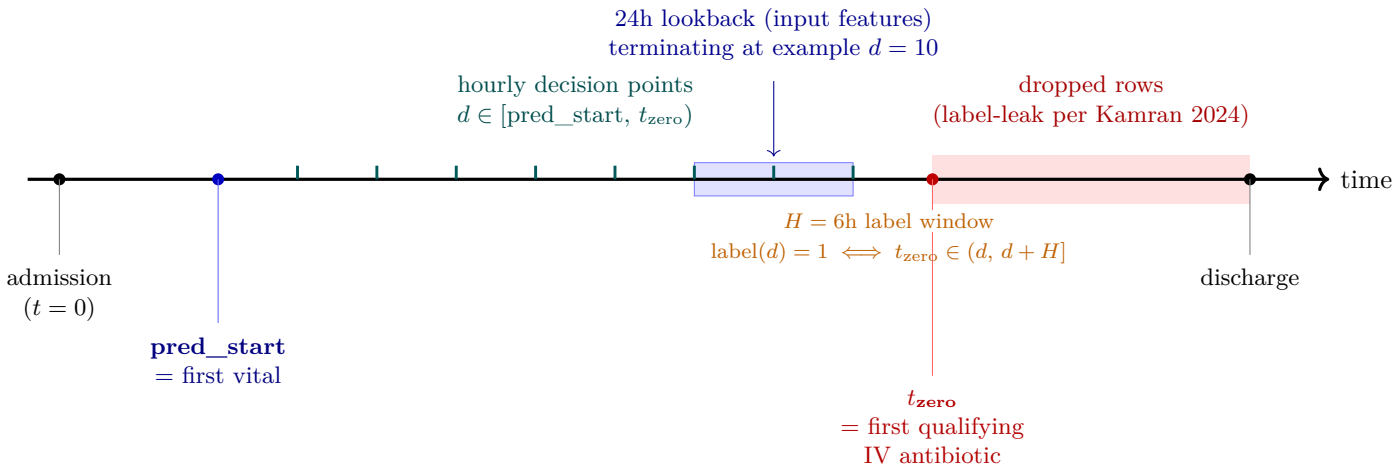


Figure 2: Hospitalization timeline. The blue "lookback" bar shows the 24h rolling feature window terminating at one example decision point. The red "dropped rows" zone covers $[t_{\text{zero}}, \text{discharge})$ — any prediction made at or after the antibiotic order is removed from training and evaluation, eliminating the label-leakage channel identified by Kamran et al. (2024).

6 Design choices and rationale

Why antibiotic-as-anchor (not SOFA $\Delta \geq 2$). The clinical workflow this task targets is the moment a clinician decides to treat for suspected infection. SOFA-anchored labels (Singer et al.,

2016; Reyna et al., 2020) describe physiology, not action: a model that predicts SOFA deterioration may be operationally useless if the deterioration is recognized faster than the model's lead time. Antibiotic order is the action we want to bring forward.

Why strictly pred_stop = t_{zero} with no Δ-gate. Kamran et al. (2024) do not introduce a fixed buffer; they restrict analysis to "predictions occurring before sepsis criteria were met and before the first indicator of treatment." Adding a buffer is an additional design choice and is documented in Table 1 for methods that adopt one. FLAIR follows the parsimonious Kamran rule: the order timestamp itself is the cutoff.

Why hospital-wide (not ICU-only). Hospital-wide matches Kamran et al. (2024)'s cohort and captures the operationally important ward-onset sepsis cases, where prediction lead time has the most clinical value. The ICU-only variant is preserved as a sensitivity arm for comparability with Shashikumar et al. (2021a) and Desautels et al. (2016).

Why hourly grid. Matches the FLAIR convention (task5 AKI-24h uses the same hourly grid) and the PhysioNet/CinC 2019 standard. ESM's native 20-minute cadence (Wong et al., 2021; Kamran et al., 2024) is finer than necessary for benchmarking and would inflate row counts ∼3× without adding label resolution.

Prevalence implications. Hospital-wide hourly grids produce highly imbalanced labels: a 7-day stay with one antibiotic event produces ≤ 6 positive rows out of ∼160. Reporting AUPRC alongside AUROC is mandatory, per the `META.evaluation_metrics` convention.

Drug list. The systemic-IV-antibiotic curation is the largest open dependency and is deferred to a follow-up round. CLIF's `med_category` mapping must be extended with an `antibiotic_systemic_iv` class. The list should follow Rhee et al. (2019)'s "qualifying antimicrobial therapy" definition (CDC ASE Appendix).

7 Out of scope

- Implementation of `flair_benchmark/tasks/task4_sepsis_abx_6h.py`.
- Curated antibiotic drug list and `med_category` mapping.
- Choice of input features and feature engineering.
- Sensitivity-arm output schemas.

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